Sibei Xia, Ph.D.

Assistant Professor

Department of Textiles, Apparel Design, and Merchandising, Louisiana State University

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https://orcid.org/0000-0002-5666-4127

EDUCATION

Doctor of Philosophy in Textile Technology Management *December 2018*

North Carolina State University, Raleigh, NC

- Dissertation: Xia, S. (2018). Predict 3D Body Measurements with 2D Images [Ph.D., North Carolina State University]. https://www.proquest.com/docview/2381675247/abstract/12DF881F86DA4606PQ/1
- Committee: Dr. Cynthia Istook, Dr. Andre West, Dr. David Bruner, and Dr. Raju Vatsavai

Master of Computer Science in Data Science Track *May 2017*

North Carolina State University, Raleigh, NC

Master of Science in Textiles *December 2013*

North Carolina State University, Raleigh, NC

- Thesis: Xia, S. (2013). *Sizing Systems Created Using SizeUSA Data for Three Body Shapes* [Master, North Carolina State University]. http://www.lib.ncsu.edu/resolver/1840.16/9807
- Committee: Dr. Cynthia Istook, Dr. Lori Rothenberg, Dr. Trevor Little, and Dr. Yingjiao Xu

Bachelor of Engineering in Fashion Design and Engineering *July 2011*

Donghua University, Shanghai, China

- Undergraduate Thesis: Design of Cup Inner Shape Based on Breast Forms
- Advisor: Dr. Fang Fang

Minor in Japanese *July 2011*

East China Normal University, Shanghai, China

RESEARCH INTERESTS

Increase customer satisfaction through product development and innovation (function-oriented, fit-oriented, and aesthetics-oriented), emerging textile and apparel technology (digital fashion and advanced knitting), and fashion informatics (text mining and computer vision)

JOURNAL ARTICLES

Xia, S., Li, J., Istook, C. L., West, A., & Bruner, D. (2023). *A Regression waist level defined for 3D body scans*. Manuscript in review.

Liu, C., **Xia, S.**, &Lang, C. (2023). Consumer coping with Covid-19: an exploratory study of clothing consumption shifts and the effect of consumer resilience. *Journal of Fashion Marketing and Management*, Vol. ahead-of-print No. ahead-of-print. https://doi.org/10.1108/JFMM-10-2022-0221

ul Hasan, H. M. R., Lang, C., & **Xia, S.** (2023). Investigating Consumer Values of Secondhand Fashion Consumption in the Mass Market vs. Luxury Market: A Text-Mining Approach. *Sustainability*, 15(1), Article 1. https://doi.org/10.3390/su15010254

Xia, S., & Liu, C. (2022). Applying Machine Learning to Study the Marketing Mix's Effectiveness in a Social Marketing Context: Fashion Brands' Twitter Activities in the Pandemic. *International Journal of Business Analytics (IJBAN)*, 9(6), 1–17. https://doi.org/10.4018/IJBAN.313416

Xia, S., Li, J., Istook, C.L. and West, A.J. (2022). A 2D image body measurement system developed with neural networks and a color-coded measurement garment. *International Journal of Clothing Science and Technology*. Vol. ahead-of-print No. ahead-of-print. https://doi.org/10.1108/IJCST-08-2021-0114

Li, J., **Xia, S.**, West, A. J., & Istook, C. L. (2022). Fashionable sportswear working as a body measurement collecting tool. *International Journal of Clothing Science and Technology*, *34*(4), 589–604. https://doi.org/10.1108/IJCST-07-2021-0090

Liu, C., **Xia, S.**, & Lang, C. (2021). Clothing consumption during the Covid-19 pandemic: Evidence from mining tweets. *Clothing and Textiles Research Journal*, 39(4), 314–330. https://doi.org/10.1177/0887302X211014973

Lang, C., **Xia, S.**, & Liu, C. (2020). Style and fit customization: A web content mining approach to evaluate online mass customization experiences. *Journal of Fashion Marketing and Management: An International Journal*, 25(2), 224–241. https://doi.org/10.1108/JFMM-12-2019-0288

Xia, S., Guo, S., Li, J., & Istook, C. (2019). Comparison of different body measurement techniques: 3D stationary scanner, 3D handheld scanner, and tape measurement. *The Journal of The Textile Institute*, *110*(8), 1103–1113. https://doi.org/10.1080/00405000.2018.1541437

Xia, S., & Istook, C. (2017). A method to create body sizing systems. *Clothing and Textiles Research Journal*, *35*(4), 235–248. https://doi.org/10.1177/0887302X17713298

Xia, S. (2015). Market forecasting of super strength fiber: Glass fiber. *Journal of Textile and Apparel, Technology and Management*, *9*(2). https://ojs.cnr.ncsu.edu/index.php/JTATM/article/view/7533

PATENTS

Istook, C. L., West, A. J., **Xia, S.**, Li, J., & Collier, J. B. H. (2020). *Devices and methods for extracting body measurements from 2d images* (United States Patent No. US20200367590A1). https://patents.google.com/patent/US20200367590A1/en

CONFERENCES

Hakeem, F., & **Xia, S.** (2022a). The fragrance of the east. *International Textile and Apparel Association Annual Conference Proceedings*, 78(1). https://doi.org/10.31274/itaa.13685

Hakeem, F., & **Xia, S.** (2022b). Beyond Abaya. *International Textile and Apparel Association Annual Conference Proceedings*, 79(1). https://doi.org/10.31274/itaa.15886

Liu, C., **Xia, S.**, & Lang, C. (2022). Coping with COVID-19: How resilient consumers clothing consumption during the pandemic. *International Textile and Apparel Association Annual Conference Proceedings*, 78(1). doi: https://doi.org/10.31274/itaa.13561

Xia, S., Li, J., Istook, C. L., West, A. (2020). What is the most appropriate way to define a 3D waist level?. *International Textile and Apparel Association Annual Conference Proceedings* 77(1). doi: https://doi.org/10.31274/itaa.11834

Xia, S. (2020). A pilot study on AutoPatternmaker: a web-based application generating customized basic block automatically. *International Textile and Apparel Association Annual Conference Proceedings* 77(1). doi: https://doi.org/10.31274/itaa.11835

Xia, S., West, A. J., Istook, C., & Li, J. (2018). Acquiring accurate body measurements on a smartphone from supplied colored garments for online apparel purchasing platforms and e-retailers. *Proceedings of 3DBODY.TECH 2018 - 9th International Conference and Exhibition on 3D Body Scanning and Processing Technologies, Lugano, Switzerland, 16-17 Oct. 2018*. https://doi.org/10.15221/18.126

Xia, S., Hada, M., Yan, J., & Nagle, H. T. (2018). Development of a posture detector using a flex sensor. *International Textile and Apparel Association Annual Conference Proceedings*, *75*(1). https://www.iastatedigitalpress.com/itaa/article/id/1381/

Xia, S., Istook, C. L., West, A., & Li, J. (2018). Algorithms behind a body measurement app built on smartphones from supplied color-coded garments for online apparel purchasing platforms. *International Textile and*

Apparel Association Annual Conference Proceedings, *75*(1). https://www.iastatedigitalpress.com/itaa/article/id/1187/

Xia, S., Guo, S., Li, J., & Istook, C. L. (2017). Comparison of body measuring techniques: whole body scanner, handheld scanner, and tape measure. *International Textile and Apparel Association Annual Conference Proceedings*, *74*(1). https://www.iastatedigitalpress.com/itaa/article/id/1904/

MEDIA INTERVIEWS

Dudgeon, C. (2023, February 15). *Smart wearable to track temperature of infants*. WTiN. https://www.wtin.com/article/2023/february/13-02-23/smart-wearable-to-track-temperature-of-infants/?freeviewlinkid=157867

Wilkins, J. (2023, January 28). *LSU researcher develops color-changing hats to detect fevers in infants; special yarn used*. The Advocate. https://www.theadvocate.com/baton_rouge/news/education/lsu-researcher-develops-hats-to-detect-infant-fevers/article_dea74eb0-9cd5-11ed-97dc-03d6bb9403cb.html

Lawrence, S. (2023, January 23). *LSU researcher develops fabric that can detect fevers in infants*. WBRZ. https://www.wbrz.com/news/lsu-researcher-develops-fabric-that-can-detect-fevers-in-infants/

Lang, M. (2023, January 20). *LSU researcher develops smart textile that detects fevers in infants*. LSU AgCenter. https://www.lsuagcenter.com/profiles/tblanchard/articles/page1674228409008

Blanchard, T. (2021, March 23). *Yoga Pants for the Win: LSU AgCenter researchers study fashion clothing consumption during COVID-19 pandemic*. LSU AgCenter. https://www.lsuagcenter.com/profiles/lbenedict/articles/page1616516372856

PROJECT IN PROGRESS

Lang, C., Su, J., & **Xia, S.** (2023). *The Effectiveness of Social Media Communication on Promoting Fashion Sustainability: A Data-Mining Approach*. Manuscript in preparation.

Xia, S. (2023). *Fever Monitor Hat Knitted with Thermochromic Yarns*. Project in progress.

Xia, S., Ruan, Y., & Gu, L. (2023). *Web Interfaced Customized Pattern Generators*. Project in progress.

Xia, S. & Li, J. (2023). *Review of Clothing Style Analysis through Computer Vision.* Project in progress.

Li, J. & **Xia, S.** (2023). S*ilhouette Detection with Neural Networks and Random Forest.* Project in progress.

Ruan, Y., **Xia, S.**, & Gu, L. (2023). *Sentiment Analysis on Consumers' Coping with Shanghai's Covid-19 Outbreak and Lockdowns Using Weibo Data*. Project in progress.

Lang, C., Liu, C., & **Xia, S.** (2023). *Resale and Consumer Resilience*. Project in preparation.

TEACHING EXPERIENCE

Assistant Professor *Fall 2019 – Present*

Dept. of Textiles, Apparel Design and Merchandising

Louisiana State University, Baton Rouge, LA

- Fall 2019
  - ✓ TAM 3037 Intermediate Apparel Product Design (3 credits)
- Spring 2020
  - ✓ TAM 3032 Textile and Apparel Product Development (3 credits)
  - ✓ TAM 3230 Pattern Design with Computer Application (3 credits)
- Fall 2020
  - ✓ TAM 3037 Intermediate Apparel Product Design (3 credits)
  - ✓ TAM 3091 Emerging Technology (3 credits)
  - ✓ TAM 7030 Creativity In Design (3 credits)
  - ✓ TAM 7091/9091 Independent Research (1-6 credits)
  - ✓ TAM 8000/9000 Thesis/Dissertation Research (1-6 credits)
- Spring 2021

- ✓ TAM 3032 Textile and Apparel Product Development (3 credits)
- ✓ TAM 3230 Pattern Design with Computer Application (3 credits)
- ✓ TAM 8000/9000 Thesis/Dissertation Research (1-6 credits)

- Fall 2021
  - ✓ TAM 3430 Emerging Technology (3 credits)
  - ✓ TAM 4037 Apparel Design Through Draping (3 credits)
  - ✓ TAM 7030 Creativity In Design (3 credits)
  - ✓ TAM 7091/9091 Independent Research (1-6 credits)
  - ✓ TAM 8000/9000 Thesis/Dissertation Research (1-6 credits)
- Spring 2022
  - ✓ TAM 1232 Digital Illustration for Apparel Designers (3 credits)
  - ✓ TAM 3032 Textile and Apparel Product Development (3 credits)
  - ✓ TAM 3230 Pattern Design with Computer Application (3 credits)
  - ✓ TAM 7091/9091 Independent Research (1-6 credits)
  - ✓ TAM 8000/9000 Thesis/Dissertation Research (1-6 credits)
- Fall 2022
  - ✓ TAM 3430 Emerging Technology (3 credits)
  - ✓ TAM 7030 Creativity In Design (3 credits)
  - ✓ TAM 7041 Introduction to Research in Textiles, Apparel Design, and Merchandising
  - ✓ TAM 8000/9000 Thesis/Dissertation Research (1-6 credits)
- Spring 2023
  - ✓ TAM 1232 Digital Illustration for Apparel Designers (3 credits)
  - ✓ TAM 3230 Pattern Design with Computer Application (3 credits)
  - ✓ TAM 8000/9000 Thesis/Dissertation Research (1-6 credits)

Graduate Lecturer *2015 – 2017*

Wilson College of Textiles, North Carolina State University, Raleigh, NC

- Fall 2015
  - ✓ FTD 215 Fashion Design 1 (3 credits)
- Fall 2016
  - ✓ FTM 219 Fashion Product Analysis Lab (3 credits)
- Fall 2017
  - ✓ FTD 215 Fashion Design 1 (3 credits)

Summer Camp Group Leader *Summer 2017*

Wilson College of Textiles, North Carolina State University, Raleigh, NC

- Teach high school students how to use their creative skills and recycle garments into newly constructed outfits

Graduate Teaching Assistant *Spring 2016 & Spring 2017*

Wilson College of Textiles, North Carolina State University, Raleigh, NC

- FTM 497 Senior Fashion Collection Studio (6 credits) & FTD 479 Senior Textile Design Studio (6 credits)
  - ✓ Assist students in preparing items for critiques and showcases
  - ✓ Assist with the preparation of items for shows

Instructor, Sewing and Embroidery Workshops *Fall 2012 – Spring 2013*

TechShop, Raleigh, NC

- Train members on using sewing machines
- Train members on using embroidery machines

Lab Assistant, Apparel Lab *Fall 2011 – Spring 2013*

Wilson College of Textiles, North Carolina State University, Raleigh, NC

- Maintain machines and other equipment in the Apparel Lab
- Assist classes that utilize the Apparel Lab

- Safety train students on using industrial sewing machines and the Lectra Cutter

STUDENT ADVISING

Committee Co-Chair (Doctoral) *Fall 2020 – present*
TAM, Louisiana State University, Baton Rouge, LA
- Student name: Fatimah Hakeem
- Research title: The ability of 3D simulation technology to produce complex garments

Committee Co-Chair (Doctoral) *2023*
TAM, Louisiana State University, Baton Rouge, LA
- Student name: Rakib Hasan
- Research title: Consumer experience of second-hand fashion consumer

Committee Co-Chair (Doctoral) *2022*
TAM, Louisiana State University, Baton Rouge, LA
- Student name: Nakib Hasan
- Research title: Understanding Consumers' Use Experience on Electrically Heated Jacket: A Study on Online Review Using Topic Modeling

Committee Member (Doctoral) *2022*
TAM, Louisiana State University, Baton Rouge, LA
- Student name: Jeremy Bernardoni
- Research title: The Prime Aesthetic: Exploratory Research on the Relationship between the Imprinting of Aesthetic Dress Preferences and the Sensitive Period of Emerging Adulthood

Committee Member (Doctoral) *2022*
TAM, Louisiana State University, Baton Rouge, LA
- Student name: Divya Desai
- Research title: Transitioning from Traditional In-Person Studio-Based Education of Apparel Design Courses to Online Education During the Covid-19 Pandemic

Committee Member (Master Non-Thesis) *2021*
TAM, Louisiana State University, Baton Rouge, LA
- Student name: Katherine Frost
- Research title: History Inspired Knitting Design

Committee Member (Master) *2020*
TAM, Louisiana State University, Baton Rouge, LA
- Student name: Saiful Islam
- Research title: Consumers' Preferred Body Scanning Technology: A Comparison

PROFESSIONAL ASSOCIATIONS

Member of the International Textile and Apparel Association (ITAA) since 2013

REVIEW & JURYING EXPERIENCE

International Textile and Apparel Association	Student best paper review in May 2022
2022 ACRA Annual Conference	Track Chair in March 2022
Textile Research Journal	Manuscript review since 2020
Clothing and Textiles Research Journal	Manuscript review since 2020
Fashion and Textiles	Manuscript review since 2020
International Textile and Apparel Association	Design exhibition jurying in June 2020
Family & Consumer Sciences Research Journal	Manuscript review since 2019

FUNDED PROJECTS

Provost's Fund for Innovation in Research Grants - Emerging Research *2023*

- Title: Wearable Technology Development with Emerging Knitting Technology: A Pilot Study of Making Baby Thermal Monitor Hats
- Principal Investigator: Sibei Xia
- Amount: $9,998

National Science Foundation *2022-2025*

- Title: Collaborative Research: FW-HTF-R: Wearable Safety Sensing and Assistive Worker-Robot Collaboration for an Augmented
- Principal Investigator: Chao Wang
- Co-principle Investigator: Fereydoun Aghazadeh, Tracey Rizzuto, and Casey Stannard
- Credit Recipient: Sibei Xia
- Amount: $720,000

Cotton Incorporated *2020*

- Title: Cotton Technologies and Innovations- Applied Learning Activity of Cotton for students in Textiles, Apparel Design, and Merchandising
- Principal Investigator: Chunmin Lang
- Co-principle Investigators: Chuanlan Liu and Sibei Xia
- Amount: $48,675

National Science Foundation (NSF) I-Corps *2017*

- Title: FitApp
- Principal Investigator: Cynthia Istook
- Entrepreneurial Leaders: Sibei Xia and Jiayin Li
- Amount: $3,000

SUBMITTED PROPOSALS

Provost's Fund for Innovation in Research - Summer Stipend *2023*

- Title: Clothing Analysis Web App Developed with Computer Vision and Fashion Informatics: A Soft-Landing Solution for Designers
- Principal Investigator: Sibei Xia
- Amount: $6,000
- Status: Pending

LA Board of Regents *2022*

- Title: Enhance Small Business Entrepreneurship in Wearable Technology with the Support of a Thermal and Sweating Manikin
- Principal Investigator: Sibei Xia
- Co-principle Investigator: Chuanlan Liu, Bruce Cameron, and Ye-Sho Chen
- Amount: $200,000
- Status: Pending

LA Board of Regents *2022*

- Title: Promoting Industrial Hemp Economy: Exploring the Markets for Hemp Fiber Textiles and Clothing
- Principal Investigator: Chuanlan Liu
- Co-principle Investigator: Sibei Xia
- Amount: $19,999
- Status: Pending

Cotton Incorporated *2022*

- Title: Fashion Entrepreneurship in Action: New Course Development and Implementation
- Principal Investigator: Chuanlan Liu

- Co-principle Investigator: Sibei Xia
- Amount: $9,963
- Status: Not Funded

Cotton Incorporated *2022*

- Title: Examining co-design entrepreneurship ideas through developing cotton aesthetic and functional knitted products
- Principal Investigator: Sibei Xia
- Co-principle Investigator: Chuanlan Liu
- Amount: $7,429
- Status: Not Funded

LA Board of Regents *2021*

- Title: Enhancing Research in Wearable Technologies Using a Thermal and Sweating Manikin
- Principal Investigator: Sibei Xia
- Co-principle Investigator: Chuanlan Liu and Bruce Cameron
- Amount: $199,892
- Status: Not Funded

US Department of Agriculture *2021*

- Title: Transforming the Capacity of Higher Education in Human Sciences to Cultivate Entrepreneurs and Promote Rural Economic Development through Embracing Digitalization and Automation Technology
- Principal Investigator: Chuanlan Liu
- Co-principle Investigator: Sibei Xia
- Amount: $150,000
- Status: Not Funded

National Endowment for the Humanities *2021*

- Title: Digital Knitting Institute: Stimulating Historical Knitting with Emerging Technologies
- Principal Investigator: Sibei Xia
- Co-principle Investigator: Chuanlan Liu and Casey Stannard
- Amount: $128,166
- Status: Not Funded

University of Alberta *2019*

- Title: Motivations and Skills to be Successful Fashion Entrepreneurs: A Longitudinal Study to Develop the Effective Fashion Business Incubation Service
- Principal Investigator: Chunmin Lang
- Co-principle Investigator: Sibei Xia
- Amount: $25,000
- Status: Not Funded

LA Board of Regents *2019*

- Title: Enhancement of Education and Research Infrastructure in Fashion Automation by Building a Holistic Fashion Technology Lab
- Principal Investigator: Sibei Xia
- Co-principle Investigator: Chuanlan Liu, Casey Stannard, and Bruce Cameron
- Amount: $199,459
- Status: Not Funded

HONORS AND AWARDS

Graduate Student Association Travel Assistance Awards *Fall 2018*

North Carolina State University, Raleigh, NC

Best Oral Presentation Award: Supply Chain and Apparel Innovation *April 2018*

Awarded by the Textile Association of Graduate Students

Wilson College of Textiles, North Carolina State University, Raleigh, NC

Recognition for Excellence in Laboratory Teaching *March 2017*

Awarded by Graduate Student Association, North Carolina State University, Raleigh, NC

The Second Prize in the Advanced Level of Cotton First Competition *November 2012*

Awarded by Wilson College of Textiles, North Carolina State University, Raleigh, NC

Special Prize in Cotton Couture *October 2011*

Awarded by Wilson College of Textiles, North Carolina State University, Raleigh, NC

Outstanding Study Scholarships *2008, 2009, & 2010*

Awarded by Donghua University, Shanghai, China

Outstanding Cadre *2009*

Awarded by Red Cross Society, Donghua University, Shanghai, China

STUDENTS AWARDS

ITAA Award for Innovative Design Scholarship – Graduate Category *October 2021*

Awarded by International Textile and Apparel Association

- Design Title: The Fragrance of the East
- Awardee: Fatimah Hakeem
- Mentor: Sibei Xia

CERTIFICATIONS

Certificate of Accomplishment in Teaching *May 2018*

Issued by the Graduate School, North Carolina State University, Raleigh, NC

Masters Track in Data Science Certificate *May 2017*

Issued by College of Engineering, North Carolina State University, Raleigh, NC

Graduate Certificate in Consumer Textile Product Design and Development *December 2013*

Issued by North Carolina State University, Raleigh, NC

Certificate of Volunteering *June 2012*

Issued by Wilson College of Textiles, North Carolina State University, Raleigh, NC

Japanese-Language Proficiency N2 *August 2010*

Jointly administered by the Japan Foundation and the Japan Educational Exchanges and Services

INDUSTRIAL EXPERIENCE

Product Developer *Spring – Summer 2014*

Gym Rags, Raleigh, NC

- Develop women's sportswear with shaping functions

Intern *Fall 2013 – Spring 2014*

Haws USA, Apex, NC

- Prepare orders
- Monitor product inventories

Intern *Spring 2013*

Royal Park Uniforms, Burlington, NC

- Alter patterns of production in Gerber AccuMark

Intern, Department of Quality Control *Summer 2011*

The Children's Place Shanghai Office, Shanghai, China

- Check samples' qualities and organize specification sheets

Intern, Department of Product Development *Spring 2011*

K-boxing Men's Wear (Shanghai) Co. LTD, Shanghai, China

- Assist product manager with preparation of the seasonal order placement fair
- Conduct trend research

Salesperson, IORI Shop *Fall 2010*

Jiuguang Dept. Store, Shanghai, China

TECHNICAL SKILLS

Programming

- Python
- R
- JavaScript
- Java

Computer-Aided Design Programs

- Optitex
- CLO3D
- Adobe Suite
- Shima Seiki design programs
- AccuMark

Machines

- Industrial sewing machines
- Lectra cutter
- NScan pattern scanner
- Size Stream 3D body scanner
- [TC]2 3D body scanner
- Shima Seiki flatbed knitting machines
- Shima Seiki whole garment printer

Others

- SPSS

LANGUAGES

Chinese: Native

English: Professional working proficiency

Japanese: Limited working proficiency